

# MODULE 5

## SEA WAVE ENERGY

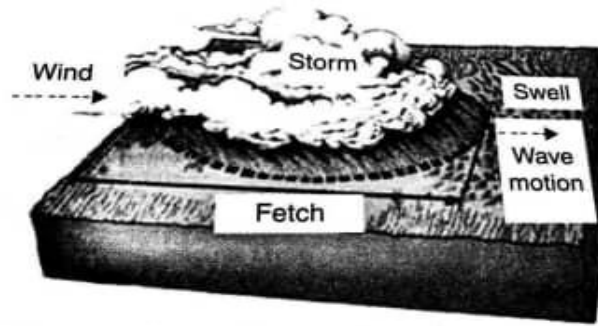
### INTRODUCTION

The energy in ocean waves mainly comes in an irregular and oscillating form at all times of the day and night. **Solar energy causes winds to blow over vast ocean areas, which in turn cause waves to form, gather, and travel huge distances to the shoreline of continents. The wave height, period, and direction are primarily dependent on the wind properties (speed, direction, and duration) and also the geometry of the sea (fetch length and depth).** There is surprisingly little loss of energy in deep-water ocean waves, so as they travel to distant shores they continue to collect more and more wind energy.

### MOTION in the Sea Waves

**When the wind blows across smooth water surface, air particles from the wind grab the water molecules they touch. Stretching of the water surface by the force or friction between the air and the water creates capillary waves.** Surface tension acts on these ripples to restore the smooth surface, and waves are formed.

The combination of forces due to the gravity, sea surface tension, and wind intensity are the main factors of origin of sea waves.



Wave speed is determined by wind speed and fetches (distance over which the wind excites the waves) and by the depth and topography of these abed (which can focus or disperse the energy of the waves). Sea waves have a regular shape at far distance from the fetch and is called swell. Wave information makes the water surface further rough and the wind continuously grips the roughened water surface and thus waves are intensified.

Waves are characterized by the following parameters

- **Crest:** The peak point (the maximum height) on the wave is called the crest.
- **Trough:** The valley point (the lowest point) on the wave is called trough.
- **Wave height (H):** It is a vertical distance between the wave crest and the next trough(m).
- **Amplitude(a):** It is defined as  $H/2$  (m).
- **Wavelength ( $\lambda$ ) :** It is the horizontal distance either between the two successive crests or troughs of the ocean waves(m).
- **Wave propagation velocity(v):** The motion of sea water in a direction(m/s).
- **Wave period(T):** It measures the size of the wave in time(s). It is the time required for two successive crests or two successive troughs to pass a point in space.
- **Frequency(f):** The number of peaks that pass a fixed point per second is defined as the frequency of wave and is given by  $f=1/T$  (cycle/sec).

## Power Associated with Sea Waves

It has been concluded by researchers through linear wave motion theory that the kinetic and potential energy ( $E$ ) of a wave per meter of crest and unit of surface can be approximated as

$$E = \rho g a^2 / 2$$

where  $\rho$  = density of water;  $g$  = gravitational acceleration; and  $a$  = amplitude of the wave (approximately equals to half its wave height  $H$ ).

The power that a meter of crest holds can be obtained by multiplying the amount of energy transported by the group velocity.

In deep water, dispersion relation ( $k$ ) is given as

$$\begin{aligned} k &= \omega^2/g, \\ \text{Further, group velocity } (V_g) &= \omega/2k = g/2\omega \\ \text{The total power } (P) \text{ is obtained as } P &= EV_g = [\rho g a^2/2](g/2\omega) = \rho g^2 a^2/4\omega \\ \text{Further, wave period } (T) &= 2\pi/\omega \text{ or } \omega = 2\pi/T \text{ and } a = H/2 \\ \text{Therefore, } P &= \rho g^2 a^2/4\omega = \rho g^2 H^2 T/32\pi \end{aligned}$$

For irregular waves of height  $H$  (m) and period  $T$ (s), an equation for power per unit of wavefront can be derived as

$$P_{\text{irregular}} = 0.4 \text{ (kW/m) of wavefront}$$

From the abovementioned equations, it is seen that the wave power is directly proportional to the square of wave height.

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### WAVE ENERGY AVAILABILITY

Density of water is about 800 times higher than air. Energy density of ocean wave are significantly several times more than air. The amount of energy available in ocean waves are tremendously high. It is considered as renewable, zero emission source of power. Estimation of the global ocean wave energy are more than 2TW according to the world energy council.

### WAVE ENERGY AVAILABILITY IN INDIA

The coastal area of Maharashtra has an annual wave potential ranging between 4kW/m and 8kW/m wave front, which is quite high as 12-20kW/m during the monsoon. Offshore Location of wave power in Maharashtra

| Site          | Average Wave Power (kW/m)<br>(Annual) | Average Wave Power (kW/m)<br>(June–August) |
|---------------|---------------------------------------|--------------------------------------------|
| Malvan Rock   | 6.91                                  | 16.73                                      |
| Kura Inset    | 5.79                                  | 13.74                                      |
| Redi          | 6.35                                  | 16.57                                      |
| Vengurla Rock | 8.01                                  | 20.61                                      |
| Square Rock   | 6.79                                  | 16.64                                      |

## DEVICES FOR HARNESSING WAVE ENERGY

There are different technologies for converting wave energy to electricity

### Terminator Devices

**Orientation:** Perpendicular to incoming waves.

**How They Work:** These devices have a stationary part and a moving part that moves up and down like a piston as waves pass. This motion pressurizes air or fluid to drive a turbine and generate electricity.

**Example:** Oscillating Water Column (OWC)

### Attenuator Devices

**Orientation:** Parallel to wave direction.

**How They Work:** Long, jointed floating structures ride the waves. The movement at the joints activates hydraulic pumps or other converters to produce electricity

### Point Absorbers

**Orientation:** No specific orientation; they absorb energy from all directions.

**How They Work:** Floating buoys move up and down with wave motion. This movement drives a mechanical system inside the device to generate electricity.

### Overtopping Devices

- **Orientation:** Typically face incoming waves directly.
- **How They Work:** Waves are guided up a ramp into a reservoir above sea level. The stored water is then released back to the ocean through turbines, generating electricity.

- **Float or Buoy devices:** Float or buoy-based wave energy converters, commonly known as **point absorbers**, are innovative devices designed to harness the energy of ocean waves. These systems consist of floating buoys anchored to the seabed, allowing them to move with the motion of the waves

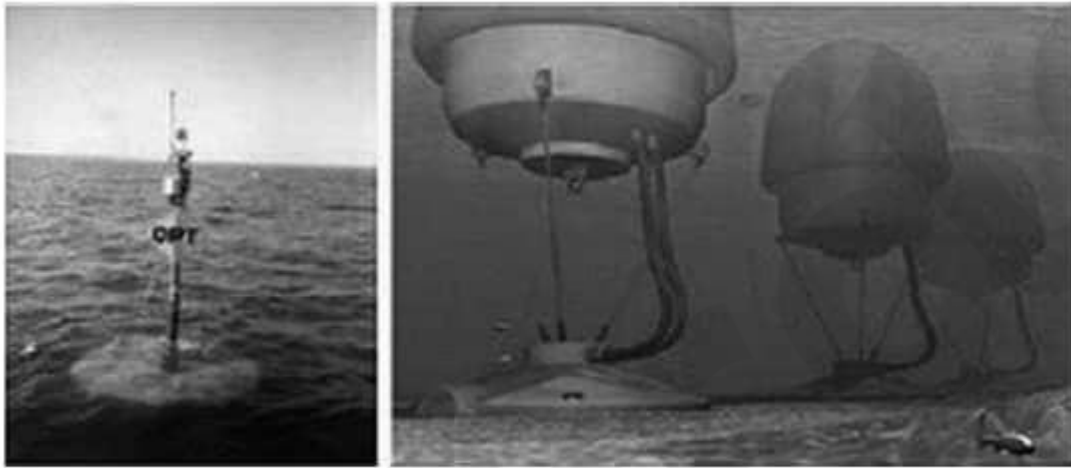
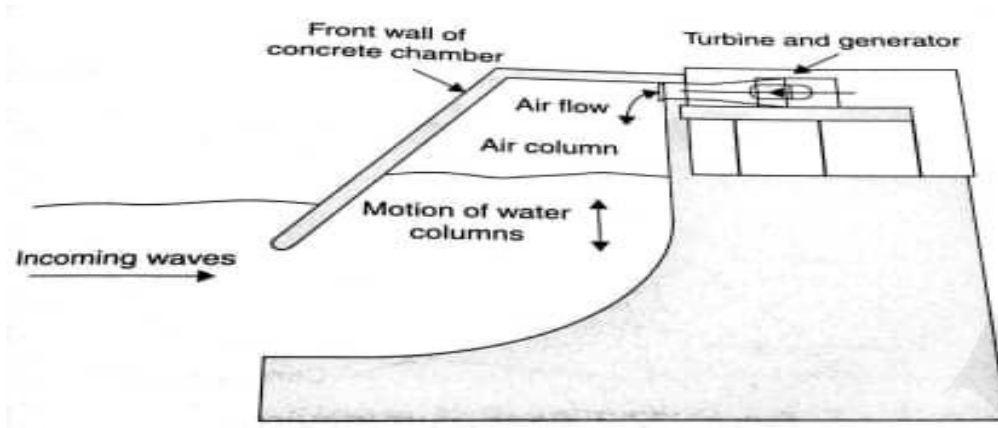


Figure 12.4 Float or buoy device

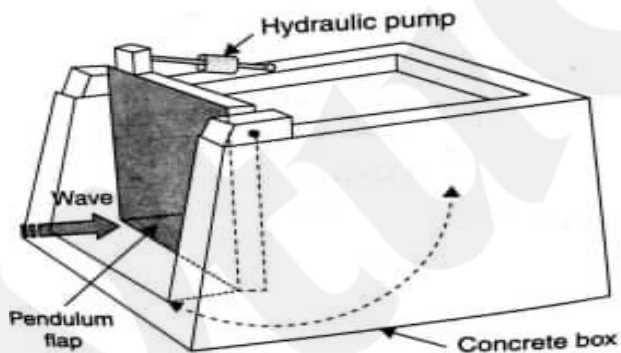
- **Oscillating water column devices:** An Oscillating Water Column (OWC) device is a shoreline-based wave energy converter that harnesses the kinetic energy of ocean waves to generate electricity. An OWC device consists of a partially submerged chamber with an opening below the waterline. As waves enter and exit this chamber, the water column inside rises and falls, compressing and decompressing the air above it. This oscillating air movement is directed through a turbine connected to a generator, converting the energy into electricity



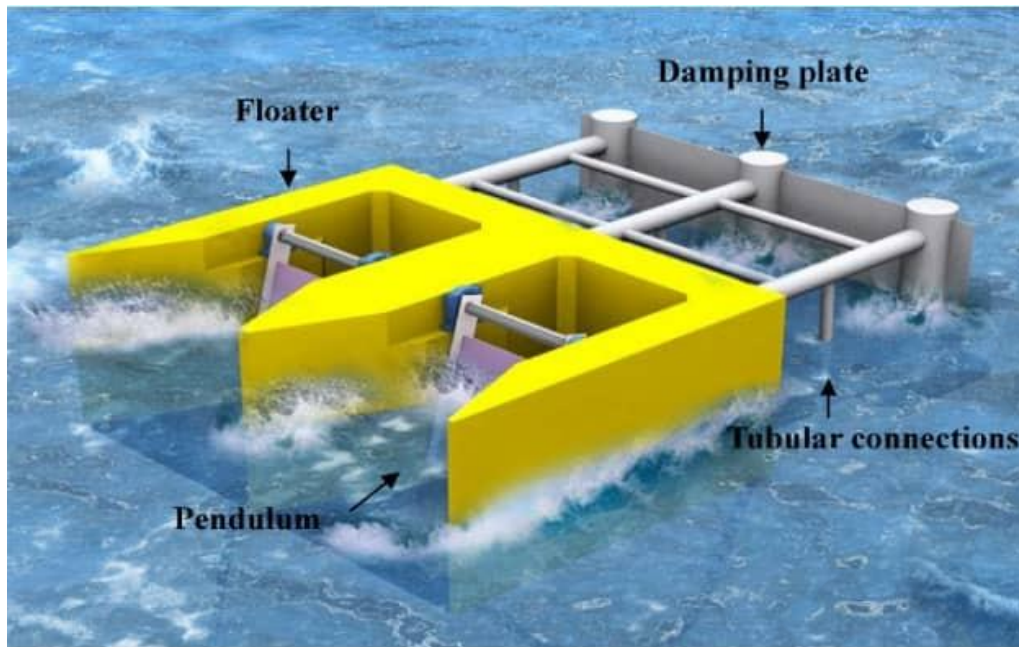
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### Pendulum systems:

It is a shoreline device that consists of a parallel piped concrete box, which is open to the sea at one end. A pendulum flap is hinged over this opening which swings back and forth by the actions of the waves. The back-and-forth motion of pendulum is then used to power a hydraulic pump and an electric generator.



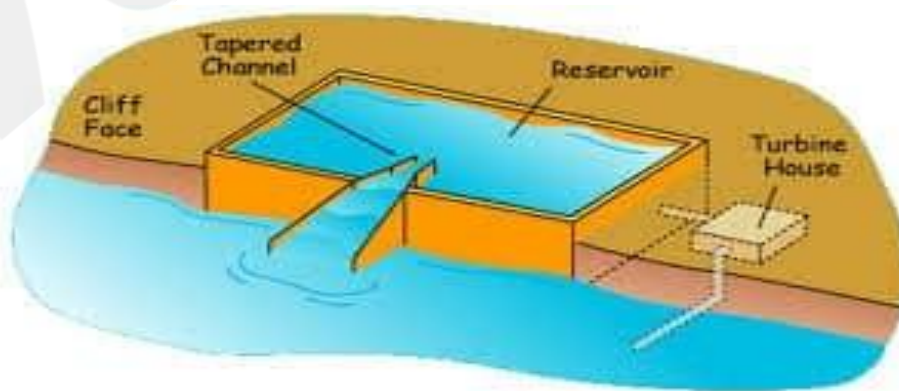




**Fig. 1** Floating pendulum wave energy converter

### **TAPCHAN (Tapered Channel):**

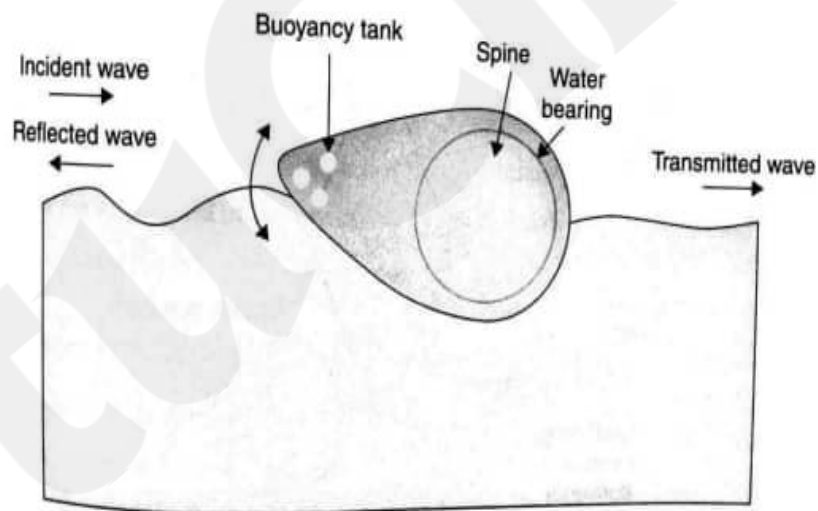
It has a tapered channel connected to a reservoir constructed above the sea level at a height of 3-5m. They are relatively low power output devices and suitable for deep water shoreline and low tidal range. It is a very simple device. Waves collect into a channel, which tapers into a large reservoir. The PE of water stored in the reservoir is extracted by releasing the reservoir water back to the sea through a low head Kaplan turbine coupled to an electrical generator.



## Salter's Duck system:

It is an egg-shaped device that moves with the motion of the waves. The shape of leading edge of the duck is in such a way that the approaching sea wave pressure is exerted on the duck. It forces the duck to rotate about a central axis. As the salter duck moves up and down on the sea waves, pendulum connected to electrical generator swings forward and backward to generate electricity. **Two sets of cables are attached to the device one to pendulum inside the device and the other to a fixed arm outside the device.**

**The cables attached to the internal pendulum contain hydraulics that pumps as the device moves back and forth with the waves. This movement of the pressurized oil pumped into hydraulic machine that drives electric generators.**

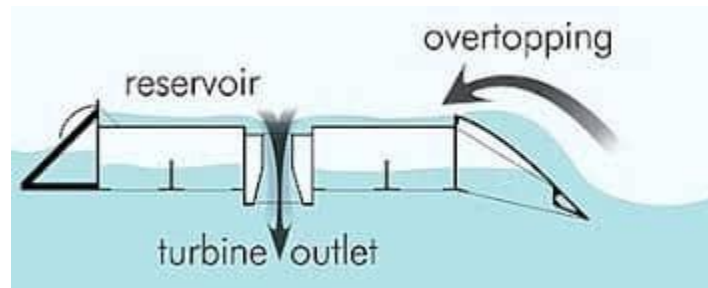






### Offshore Wave Dragon system:

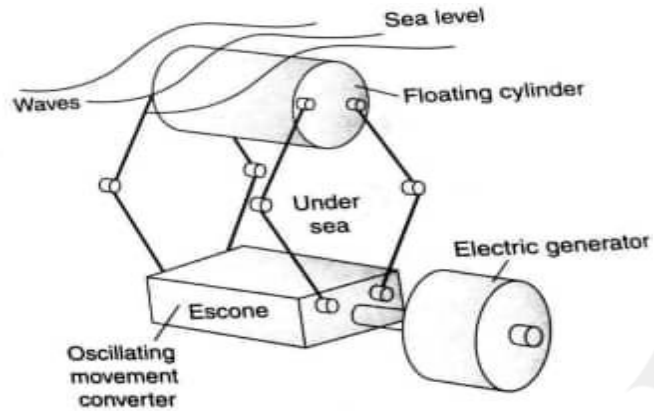
It is an overtopping device that elevates ocean waves to a reservoir above sea level. Water is let out through a number of turbines and in this way, it is transformed into electricity. It consists of two large arms that focus waves up ramp into a reservoir. The water returns to the ocean by the force of gravity via a low head hydro turbine that drives an electric generator.



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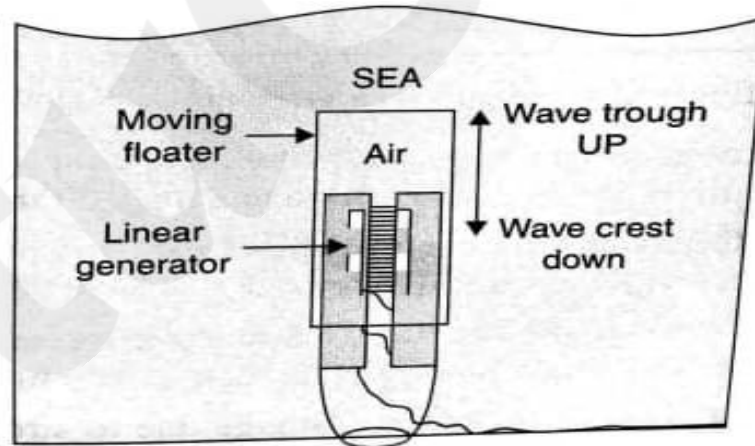
### Bristol cylinder:

It consists of a floating cylinder that collected the wave's movement. The cylinder is mechanically connected to the energy unit by flexible joints and rods. The rods are moving slowly with cylinder and the reciprocating motion is transferred to the axels in converter unit.



### Archimedes wave swing devices:

It is an underwater buoy of which the upper part moves up and down in the wave, while the lower part stays in position. The floater is pushed down under a wave crest and moves up under a wave trough. The interior of the system is pressurized with air and serves as an air spring. The mechanical power is converted into electrical power by means of a power take off system.



## ADVANTAGES AND DISADVANTAGES OF WAVE POWER

### Advantages

1. Sea waves have high energy densities and provide a consistent stream of electricity generation capacity.
2. Wave energy is clean source of renewable energy with limited negative environmental impacts.

3. It has no greenhouse gas emissions or water pollutants.
4. Operating cost is low and operating efficiency is optimal.
5. Damage to ocean shoreline is reduced.

### Disadvantages

1. High construction costs.
2. Marine life is disrupted and displaced.
3. Damage to the devices from strong storms and corrosion create problems.
4. Wave energy devices could have an effect on marine and recreation environment.

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### OCEAN THERMAL ENERGY

**Ocean Thermal Energy Conversion (OTEC)** is a technology that uses the temperature difference between warm surface water and cold deep ocean water to produce electricity.

The sun heats the surface of the ocean, especially in tropical regions between the Tropic of Cancer and Tropic of Capricorn. Meanwhile, deep ocean water stays cold. This creates a temperature difference, or **thermal gradient**.

- When the temperature difference is **20°C or more**, a large amount of electricity can be generated.
- In tropical areas, surface water is usually around **25°C**, while water 1,000 meters deep can be as cold as **5°C**.
- In colder places like the **Baltic Sea**, where the average temperature is only about 10°C, OTEC systems are not practical because the temperature difference is not big enough.

So, **OTEC works best in warm ocean areas**, and it turns the ocean's heat into a **renewable source of electricity**.

**Bonus benefit:** The cold water from deep in the ocean is full of nutrients. After it is used in the OTEC system, it can help grow plants and marine life near the coast or on land.

### Principle of OCEAN THERMAL ENERGY CONVERSION

There are **two main types** of OTEC systems: **Closed-Cycle** and **Open-Cycle**. Both use the temperature difference between warm surface water and cold deep ocean water to make electricity.

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## Closed-Cycle OTEC:

1. **Warm surface water** is pumped through a **heat exchanger**.
2. It heats a **working fluid** (like ammonia or propane), which has a **low boiling point**.
3. This fluid **boils into vapor** and builds pressure.
4. The vapor spins a **turbine**, which drives a **generator** to produce electricity.
5. The vapor then goes to a **condenser**, where **cold deep ocean water** cools it down and turns it back into a **liquid**.
6. This liquid is sent back to the heat exchanger to **repeat the cycle**.

 **Cycle continues** if the temperature difference is maintained.

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## Open-Cycle OTEC:

1. **Warm ocean water** is pumped into a **low-pressure chamber**, where it **boils into steam** naturally (without needing another fluid).
2. The steam turns a **turbine** to generate electricity.
3. Then, **cold deep water** cools the steam in a **condenser**, turning it back into **freshwater**.

 This method also gives the added benefit of **desalinated water**.

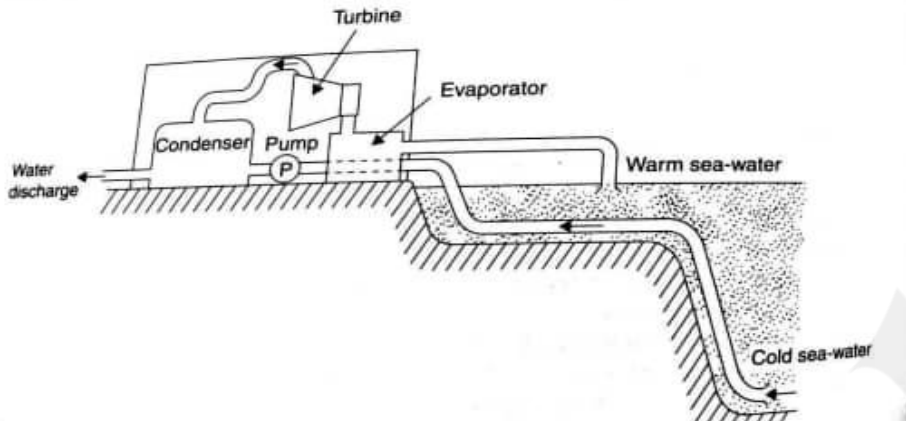
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## Energy Usage:

- A small part of the electricity made is used to **run the OTEC system** itself (like pumps and controls).
- The rest is sent to **power homes, industries, or other systems**.

## OCEAN THERMAL ENERGY CONVERSION PLANTS

Land based power plant



A **land-based OTEC power plant** is built **onshore**, close to the ocean. All the equipment needed to run the plant is placed inside a building on land.



### How It Works:

1. **Three long pipes** connect the plant to the ocean:
  - o **Warm water pipe** brings surface water from near the shore.
  - o **Cold water pipe** goes deep into the ocean to bring up cold water.
  - o **Discharge pipe** returns the used water back to the ocean, often through **marine culture ponds**, which can help grow fish or plants.
2. The **warm and cold water** are used in the OTEC process to generate electricity (like in closed-cycle or open-cycle systems).



### Challenges:

- **Very expensive** to install and maintain the long underwater pipes.
- A lot of electricity is needed just to **pump the water**, which reduces the **net electricity output**.

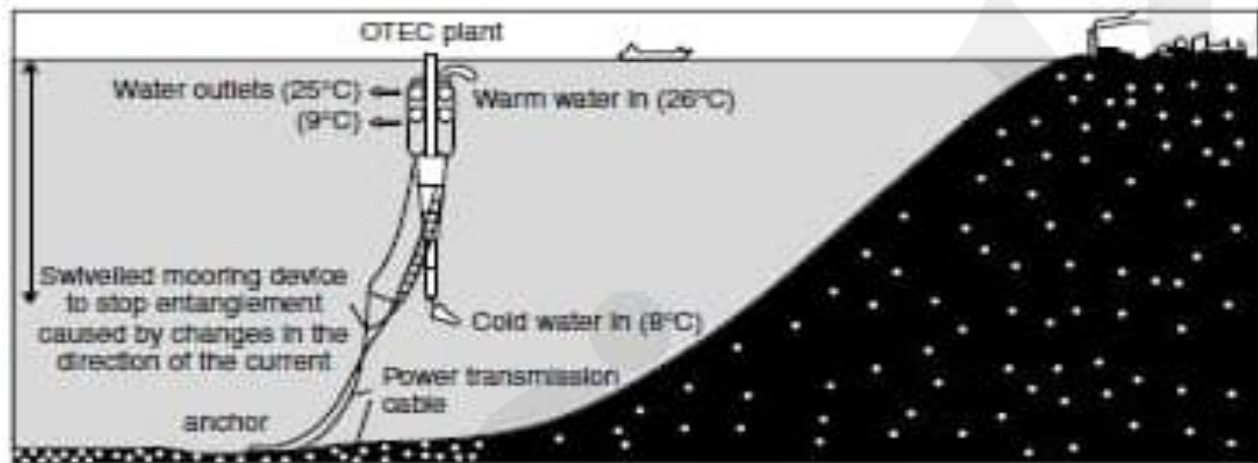


### Benefits:

- Since it is built on land, it is **easier to connect** to the existing **electric grid**, so power delivery is more efficient.
- No need for extra infrastructure to send electricity inland.

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### Floating power plant:



A **floating OTEC plant** is built on a **ship-like platform** in the ocean, right where there is a **strong temperature difference** between warm surface water and cold deep water — ideal for OTEC.

#### ⚙️ How It Works:

- It uses the **same OTEC process** (closed-cycle or open-cycle) as a land-based plant.
- Since it floats directly above the spot where warm and cold water are available, it **saves money on long water pipes**.



#### Challenges:

- Although it saves on water piping, it needs **long underwater power cables** to send electricity back to the land.
- These cables are **very expensive, inefficient**, and can have **technical issues** in deep ocean environments.





### Alternative Use:

- To avoid the problems of long-distance power transmission, **floating OTEC plants often produce fuel instead**, such as:
  - **Hydrogen**, made by using electricity from OTEC to split water into hydrogen and oxygen (**electrolysis**).
- This fuel can then be **transported to land** by ship, instead of transmitting electricity through cables.

## BASIC RANKINE CYCLE AND ITS WORKING

### Basic Rankine Cycle (Used in OTEC)

The **Rankine cycle** is a basic process used in power plants—including OTEC systems—to convert **heat into electricity**. It works in a **loop** and involves heating, expanding, cooling, and pumping a **working fluid**.

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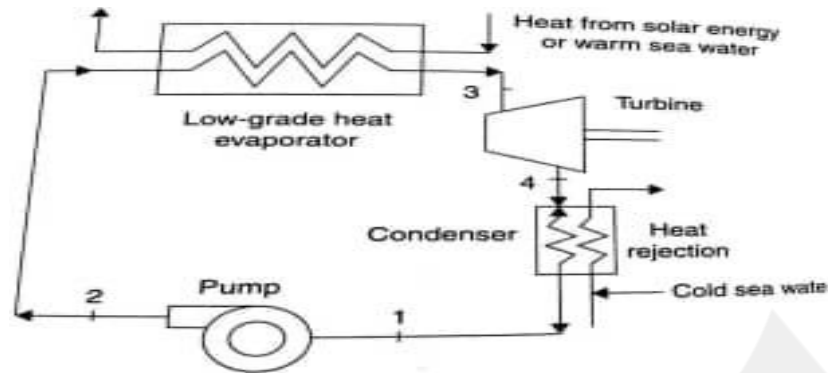


### Main Components:

1. **Evaporator** – Heats the fluid until it turns into vapor.
  2. **Turbine Expander** – Vapor flows through this and **spins the turbine**.
  3. **Condenser** – Cools the vapor back into a liquid.
  4. **Pump** – Moves the liquid back to the evaporator.
  5. **Working Fluid** – The substance that goes through the cycle (like **ammonia** or **propane** in closed-cycle OTEC; or **sea water** in open-cycle).
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### ⚙️ Step-by-Step Working:

1. **Warm ocean surface water** is the heat source.
2. The **pump** sends the working fluid (liquid) into the **evaporator**.
3. In the **evaporator**, warm water heats the fluid, which **boils into vapor**.
4. The vapor moves into the **turbine**, spinning it. The turbine is connected to a **generator**, which produces **electricity**.
5. The **vapor then flows into the condenser**.
6. **Cold deep ocean water** cools the vapor in the condenser, turning it back into a **liquid**.
7. The **cycle repeats** as the fluid is pumped back to the evaporator.



## Selection of working fluids:

OTEC uses a **Rankine cycle** to produce electricity, and the **working fluid** is what flows through the cycle to carry heat and convert it into energy.

There are two main types of Rankine cycles:

- **Steam Rankine Cycle** – uses **water** as the working fluid.
- **Organic Rankine Cycle (ORC)** – uses **organic fluids** like **ammonia, propane**, or other refrigerants.

Choosing the **right fluid** is critical for performance, safety, and efficiency.

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## Good Working Fluids Should Have These Qualities:

1. **Chemical Stability & Compatibility**
  - Fluid should not **break down** or **react** with metal parts under heat and pressure.
  - Prevents **corrosion** or **explosions**.
2. **Good Heat Transfer**
  - Fluids with **low thermal resistance** transfer heat better, improving performance.
3. **High Flash Point**
  - A high flash point means it is **less likely to catch fire**, which improves safety.
4. **Low Specific Heat**
  - Fluids with low specific heat need **less energy to change temperature**, so **cooling them is easier**, reducing the load on the condenser.
5. **High Latent Heat**
  - Fluids that absorb **more heat during boiling** are more efficient, helping **recover more energy**.
6. **Safe to Use**
  - Non-toxic, non-corrosive, and **non-flammable** fluids are safer for workers and equipment.

7. **Environmentally Friendly**

- o Fluids should not **pollute the ocean** if there is a leak—low toxicity is ideal.

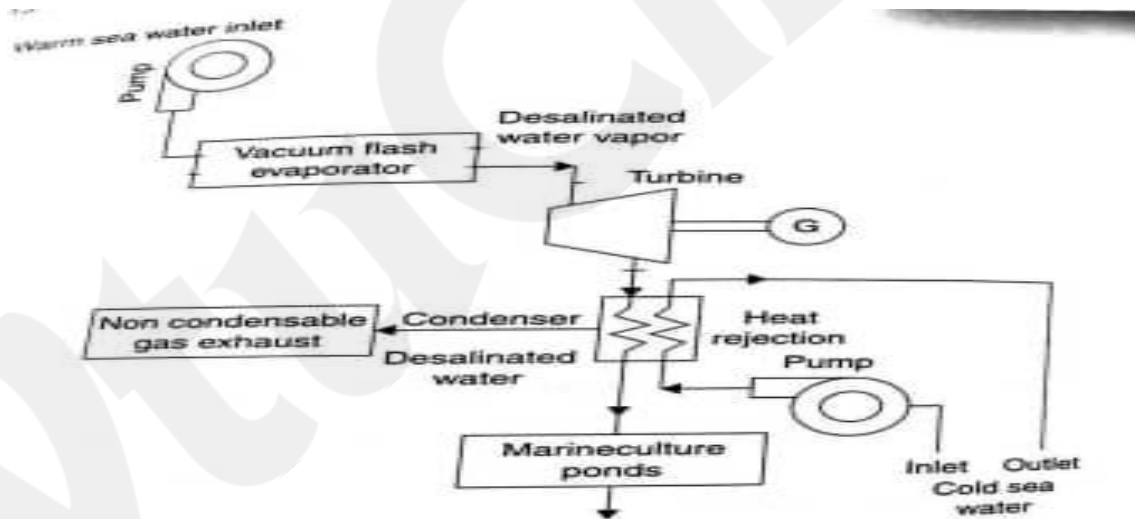
8. **Affordable and Available**

- o The fluid should be **easy to find** and **affordable**, to keep the plant economical.

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### OPEN CYCLE OTEC

An open-cycle OTEC uses the **warm ocean surface water as working fluid**. It is a non-toxic and environment friendly fluid. The major components of this system are shown in Figure



The working principles of open-cycle OTEC plants are explained as follows with the help of Figure 13.4.

1. The warm ocean surface water is pumped into flash evaporator where it is partially flashed

into steam at a very low pressure. The remaining warm sea water is discharged into the sea.

2. The low-pressure vapour (steam) expands in turbine to drive a coupled electrical generator

to produce electricity. A portion of electricity generated is consumed in plants to run pumps and for other work, and the remaining large amount of electricity is stored as net electrical power.

3. The steam with many gases (such as oxygen, nitrogen, and carbon dioxide) released from

the turbine separated from sea water in an evaporator is pumped into condenser. The steam is cooled in a condenser by cold deep sea water.

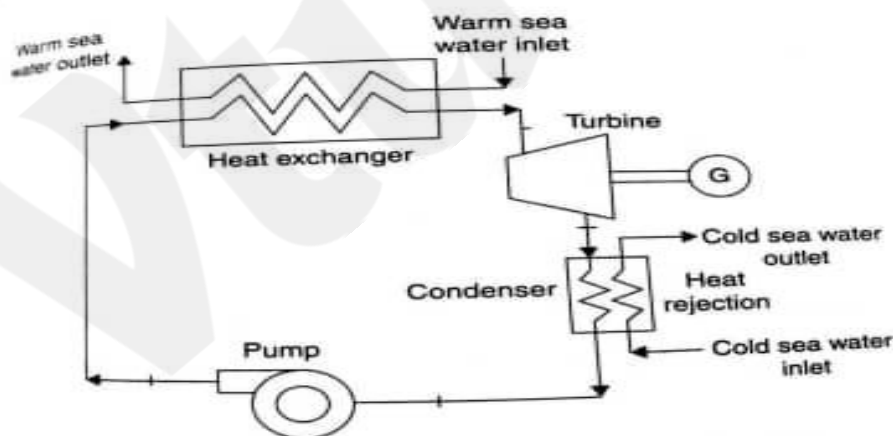
4. The condensed non-saline water is discharged either directly in deep sea cold water or through the marine culture pond.

5. The non-condensable gases are compressed to pressure and exhausted simultaneously.

6. The warm ocean surface water is continuously pumped into evaporator and cycle repeats.

## CLOSED CYCLE OTEC

The schematic of closed-cycle OTEC is shown in Figure. It has different arrangement when compared to open-cycle OTEC

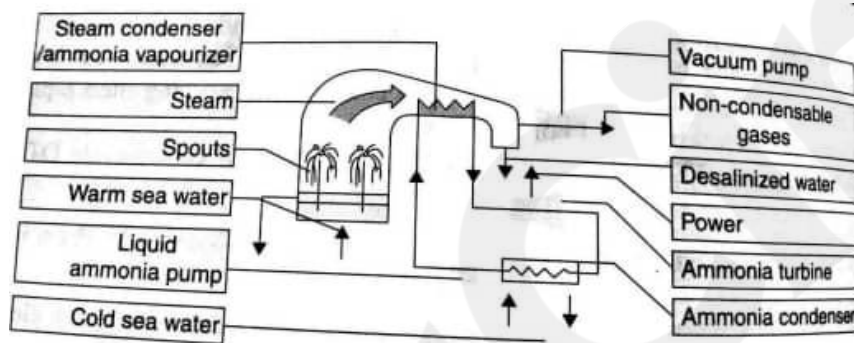


Working principles of closed-cycle OTEC are as follows:

1. Working fluid is pumped through heat exchangers in a closed loop cycle which is perfectly leakage proof.
2. Warm sea surface water is pumped through separate pipe in heat exchanger in close contact with fluid closed loop cycle

3. Warm sea water transfer its heat energy to working fluid in heat exchanger and working fluid vaporizes.
4. The fluid vapour makes the turbine to rotate and drive an electrical generator to produce electricity.
5. Fluid vapour leaving the turbine is cooled and condensed as liquid fluid and is pumped again to repeat cycle.
6. Cold deep-sea water is pumped through a separate pipe in condenser for providing efficient cooling of working fluid.

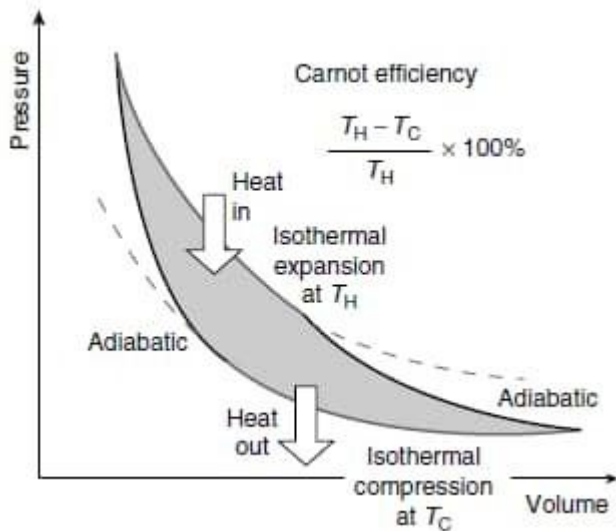
## OTEC HYBRID CYCLE



As shown in Figure, a hybrid cycle combines the features of both closed-cycle and open-cycle systems. Warm sea water is pumped into a vacuum chamber where it is used to flash and produces steam. Working fluid in another closed cycle loop is evaporated and vapourized by steam in vacuum chamber. The fluid vapour rotates the turbine and drive an electric generator to produce electricity.

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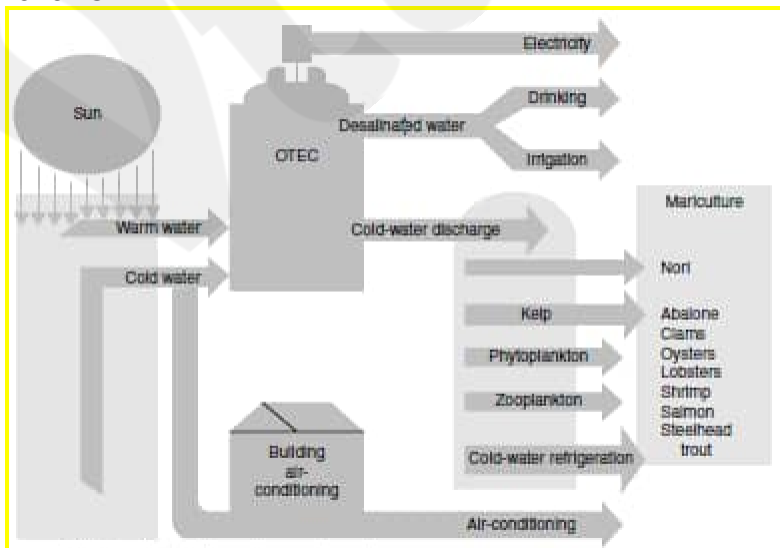
## CARNOT CYCLE



- Let  $W$  be the work done by the system
- $Q_H$  be the heat put into the system.
- $T_C$  be the absolute temp of the sea surface
- $T_H$  be the absolute temperature of the deep sea water hot reservoir.
- Carnot efficiency is given as
- $\eta = W/Q_H = 1 - T_C/T_H$

## APPLICATIONS OF OTEC IN ADDITION TO ELECTRICITY

OTEC schematic diagram and applications are shown in Figure Ocean thermal converting plants provide several products for use by mankind. These are explained as follows:



1. **Electricity:** Electrical energy is the primary product of OTEC plants. Laying down long transmission and distribution cables up to the sea shore for domestic and industrial applications is not practical from economic view point. OTEC plants are, therefore, considered for other products and applications.
2. **Hydrogen production:** Electricity produced from OTEC plants is used for separating water in hydrogen and oxygen by the method of electrolysis of water. Hydrogen is considered and OTEC electricity for hydrogen production signifies the important applications of OTEC plants.
3. **Ammonia and methanol production:** OTEC electricity can be used to obtain by-products, such as ammonia and methanol that can be transported either by tankers or through pipe lines to on shore applications.
4. **Desalinated water:** Desalinated water is produced in an open-cycle and hybrid-type OTEC plants through surface condenser. It is freshwater and widely used as water resource for drinking, agriculture, and industry.
5. **Aquaculture:** Nutrient-rich cold deep-sea water provides sufficient environment for fish farming which may create a profitable business activity.
6. **Chilled soil agriculture:** Chilled soil agriculture is another application of OTEC plants. Cold deep-sea water flowing through underground pipes chills the surrounding soil. The temperature difference is maintained between plant roots in the cool soil and plant leaves in the warm air, and thus, the tree and plants grow. The amount of food that can be produced in this way is very large, larger in market value than the electric power produced by the plant.
7. **Air conditioning:** Because the temperature is only a few degrees, cold water can be used as a fluid in air condition systems.

## ADVANTAGES, DISADVANTAGES AND BENEFITS OF OTEC

### Advantages

1. Ocean thermal energy is a renewable, clean natural resource available in abundance.
2. It is pollution-free and has no greenhouse effects.
3. It is a good source of freshwater and portable water.

### Disadvantages

1. High cost: Electricity generated by OTEC plants is more expensive than electricity produced by chemical and nuclear fuels.
2. Complexity: OTEC plants must be located where a difference of about 20°C occurs year round. Ocean depths must be available fairly close to shore-based facilities for economic operation. Floating plant ships could provide more flexibility.
3. Acceptability: For the large-scale production of electricity and other products, OTEC plants are poorly acceptable due to their high costs.
4. Ecosystem damage: It is obvious by setting OTEC plants.



5. Lower efficiency: A higher temperature difference between ocean surface warm water and cold deep ocean water is required for highly efficient operation of plant.

## Benefits as a Measure of the Value of OTEC

Economic and other benefits are the value of OTEC plants. These include the following:

1. It is a clean, renewable natural resource available in plenty.
2. It has no environmental problems and greenhouse effects.
3. It is a source of base load electricity and fuels such as hydrogen, methanol, and ammonia.
4. It provides freshwater for drinking, agriculture, and industry.
5. It encourages chilled agriculture and aquaculture.
6. Self-sufficiency, no environmental effects, and improved sanitation and nutrition are the added benefits for island.

## Questions

1. Explain motion in the sea waves.
2. Explain devices used for harnessing wave energy.
3. Explain advantages and disadvantages of wave power.
4. Explain principle of ocean thermal energy conversion.
5. Explain Ocean thermal energy conversion plants.
6. Explain basic rankine cycle and its working.
7. Explain closed cycle, open cycle and hybrid cycle OTEC.
8. Explain Carnot cycle of OTEC.
9. Explain applications of OTEC.
10. Explain advantages, disadvantages and benefits of OTEC.